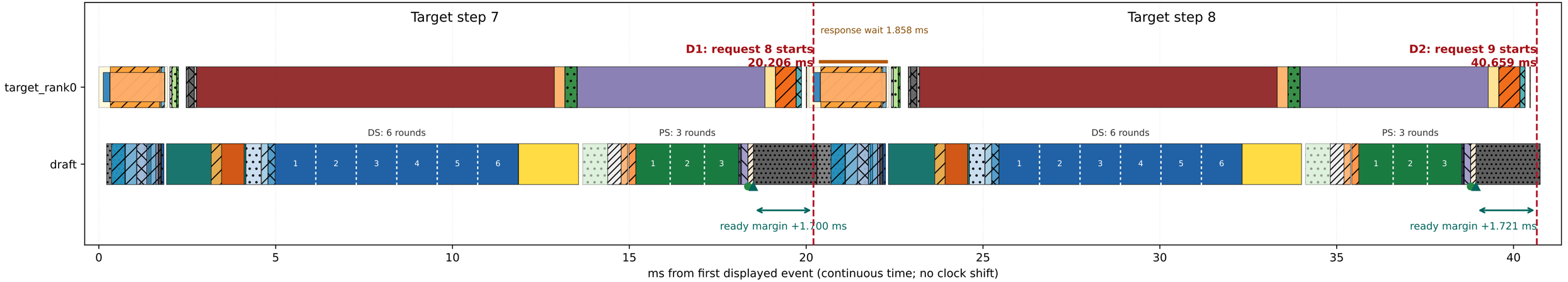
LayerSkip Llama2-7B + AMD-Llama-135M | Adopted PS optimization | P1 hit | UID 514_t0 | steps 7-8

Runtime matches ver1.2 9390e21 | RTX4090 | B1 BF16 T0.7 | exit20 K1/K2 6/3 M1/M2 12/8 C3 | frontier

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

In this measurement: no observed proxy computation / target graph_post overlap. Send enqueue does not measure transfer completion.

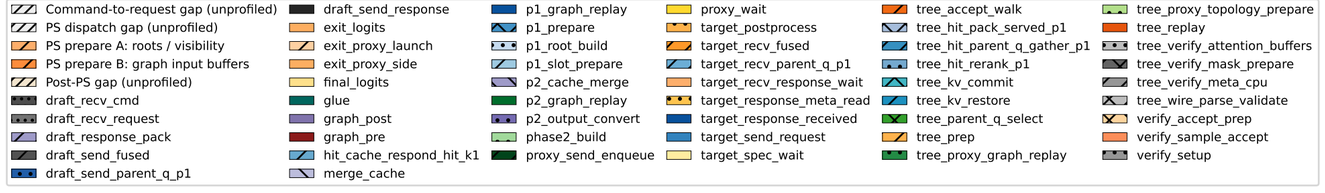


MEANS OVER ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

Current outcome	Steps	Step (ms)	DS total (ms)	DS spare* (ms)	DS graph/K1 (ms)	PS total (ms)	PS spare (ms)	PS graph/K2 (ms)
P1 hit	813	19.875	7.715	+1.529	1.137	4.914	+1.482	0.953
P2 hit	1245	19.656	7.567	+1.925	1.109	4.943	+1.391	0.953
Miss	333	20.913	6.604	+3.478	0.952	4.905	+1.638	0.947
Overall	**2391**	**19.905**	**7.483**	**+2.007**	**1.097**	**4.928**	**+1.456**	**0.952**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Positive spare = finishes early; negative = late. Graph/K is an average per forward round INCLUDING sampling/selection, not isolated transformer latency.



Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

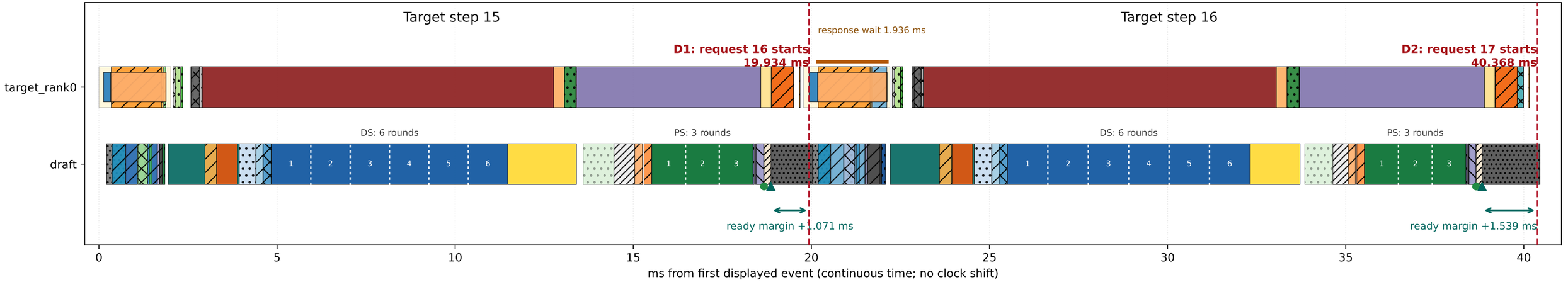
LayerSkip Llama2-7B + AMD-Llama-135M | Adopted PS optimization | P2 hit | UID 397_t0 | steps 15-16

Runtime matches ver1.2 9390e21 | RTX4090 | B1 BF16 T0.7 | exit20 K1/K2 6/3 M1/M2 12/8 C3 | frontier

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

In this measurement: no observed proxy computation / target graph_post overlap. Send enqueue does not measure transfer completion.

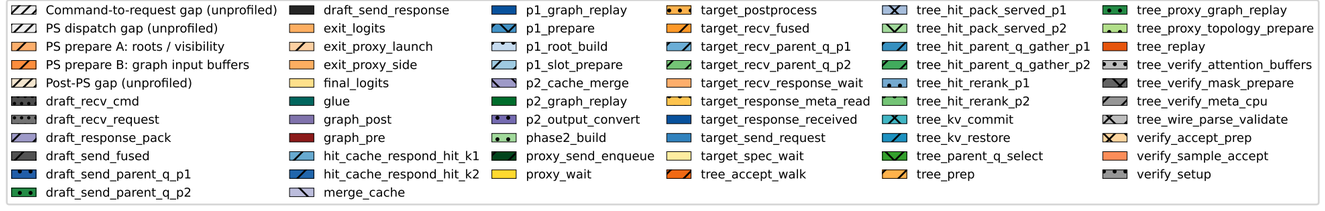


MEANS OVER ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

Current outcome	Steps	Step (ms)	DS total (ms)	DS spare* (ms)	DS graph/K1 (ms)	PS total (ms)	PS spare (ms)	PS graph/K2 (ms)
P1 hit	813	19.875	7.715	+1.529	1.137	4.914	+1.482	0.953
P2 hit	1245	19.656	7.567	+1.925	1.109	4.943	+1.391	0.953
Miss	333	20.913	6.604	+3.478	0.952	4.905	+1.638	0.947
Overall	**2391**	**19.905**	**7.483**	**+2.007**	**1.097**	**4.928**	**+1.456**	**0.952**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Positive spare = finishes early; negative = late. Graph/K is an average per forward round INCLUDING sampling/selection, not isolated transformer latency.



Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

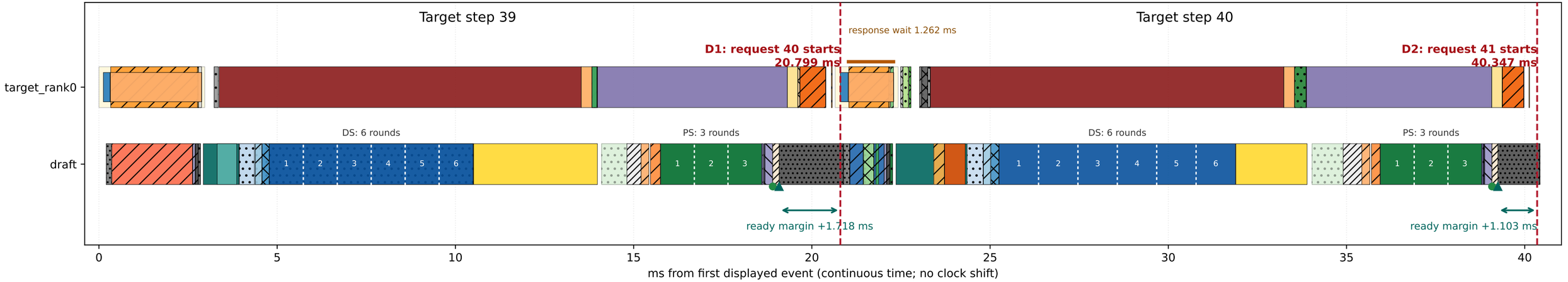
LayerSkip Llama2-7B + AMD-Llama-135M | Adopted PS optimization | Miss | UID 234_t0 | steps 39-40

Runtime matches ver1.2 9390e21 | RTX4090 | B1 BF16 T0.7 | exit20 K1/K2 6/3 M1/M2 12/8 C3 | frontier

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

In this measurement: no observed proxy computation / target graph_post overlap. Send enqueue does not measure transfer completion.

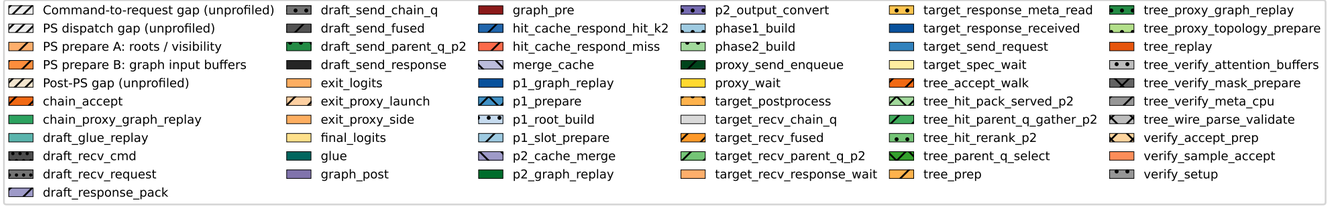


MEANS OVER ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

Current outcome	Steps	Step (ms)	DS total (ms)	DS spare* (ms)	DS graph/K1 (ms)	PS total (ms)	PS spare (ms)	PS graph/K2 (ms)
P1 hit	813	19.875	7.715	+1.529	1.137	4.914	+1.482	0.953
P2 hit	1245	19.656	7.567	+1.925	1.109	4.943	+1.391	0.953
Miss	333	20.913	6.604	+3.478	0.952	4.905	+1.638	0.947
Overall	**2391**	**19.905**	**7.483**	**+2.007**	**1.097**	**4.928**	**+1.456**	**0.952**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Positive spare = finishes early; negative = late. Graph/K is an average per forward round INCLUDING sampling/selection, not isolated transformer latency.



Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.